

An identifier hit is not always a safe match.

A catalog operator needs an explainable queue, not a guessed answer.

The buyer problem: regulator records and internal catalog rows can share an identifier while disagreeing on product strength. A confident-looking false match could send the wrong SKU into an operational workflow.

The useful result: the real matcher preserves both candidates, shows the conflicting evidence, and routes the row to a person as **REVIEW NEEDED**.

RM

PRODUCT RECALL MATCH DESK
Runnable evidence / WS-002

REAL PM

CATALOG IMPACT REVIEW

An identifier hit is evidence. It is not always a safe match.

Edit one product row, run the matcher, and inspect why it matched or stopped for review.

01 / EDIT THE CATALOG EVIDENCE

Levothyroxine catalog row

Product name

Levothyroxine Sodium 100 mcg

Manufacturer

Major Pharmaceuticals

UPC

0055154356305

NDC family

Lot code

Run the matcher

Load ambiguous row

Add decisive evidence

Recorded, allowlisted openFDA data. No new FDA request occurs. Not medical advice, a public alert, or current recall status.

Run automated checks

Checks have not run in this session.

02 / INSPECT THE DECISION

CAT-005 / REVIEW NEEDED

Python completed in 5.117 ms / 2026-09-04T03:41:40.570330+00:00 / run ea48eff2

8

Catalog rows

6

Matches

1

Review

1

No match

CAT-005 decision

REVIEW NEEDED
 Shared identifiers conflict with the declared strength. A person must verify the source candidates.

03 / FOLLOW THE EVIDENCE

Top source candidates

D-0709-2026

50 points / re

Exact UPC

Manufacturer overlap

Strong product text overlap

Strength conflicts

D-0711-2026

50 points / re

Exact UPC

Manufacturer overlap

Strong product text overlap

Strength conflicts

D-0737-2026

15 points

Manufacturer overlap

Source trail and fingerprints

Source request

https://api.fda.gov/drug/enforcement.json?search=openfda.upc:*&sort=report_date:desc&limit=1

Captured

2026-09-04T03:06:10.1206316Z

Catalog input SHA-256

06c3182d548d9ef79561d2b2d7cfd8d1974790e5181d0eda0ee91f2ed166ad5d

Actual local-browser execution on 8 synthetic catalog rows and 10 selected, recorded openFDA enforcement records. The source request, capture time and fingerprints remain visible in the workbench.

WS-002 / 1 of 3

Identity evidence has to agree by field.

The creative obstacle: a shared UPC can span strengths and package variants.

What failed in the first expansion run: broad text overlap and a shared identifier were too permissive. Only 7 of 20 independently declared benchmark labels matched the output.

The correction: compare strength values within the same unit, use an overlap coefficient for short names, and keep a visible review band. An exact identifier with a strength conflict is capped as review evidence; matching NDC, lot and strength can make one candidate decisive.

Actual evidence ladder excerpt

```
exact_upc = bool(catalog_upc and catalog_upc in recall_upcs)
strength_match = any(
    strength_conflict = any(
        score = 0
    if exact_upc:
        score += 70
        reasons.append("exact_upc")
        score += 50
        reasons.append("exact_ndc_family")
        score += 35
        reasons.append("exact_lot_code")
        score += 15
        score += 20
```

50

conflict score - review

200

clarified score - match

2

source candidates retained

0

silent conflict overrides

The repair was bounded to one pass. It changed reusable matching logic and tests; it did not hand-edit output rows or alter the independent labels.

Boundary: this is entity-resolution evidence for human catalog review. It is not a medical determination, public alert, current recall-status service or proof of source-wide coverage.

Change the evidence. Watch the decision change.

Reproduce the result, inspect provenance and export the executed run.

20/20

independent labels reproduced

21

automated tests passed

13/1/6

match / no match / review

1

recorded source request

Contrasting case: change CAT-005 from 100 mcg with no NDC or lot to 75 mcg plus NDC 55154-3560 and lot N02172A. The queue changes from 6 match / 1 review / 1 no match to 7 / 0 / 1, and CAT-005 becomes one explainable match.

02 / INSPECT THE DECISION

Executed

CAT-005 / MATCH

Python completed in 4.085 ms / 2026-09-04T03:42:11.513686+00:00 / run c5faba7a

8

Catalog rows

7

Matches

0

Review

1

No match

CAT-005 decision**MATCH**

Matched D-0709-2026 with a decisive evidence combination.

03 / FOLLOW THE EVIDENCE

Every reason is deterministic

Top source candidates**D-0709-2026**

200 points / match

Try it locally in about one minute

1. In the project folder, run `python -B src/demo_server.py`.
2. Open the local link below and click Run the matcher.
3. Click Add decisive evidence, rerun, and compare the reasons and input hash.
4. Download JSON/CSV and click Run automated checks.

Local working demo: 127.0.0.1:8766**Repository, data, evidence and launch instructions**

20 synthetic benchmark rows reproduced 20 independently declared labels: 13 match, 1 no match and 6 review-needed. All 21 automated tests passed. A separate non-builder validation and public signed-out check are required before release.